AUTOMATION & CONTROL INSTITUTE
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Multi-log forestry crane

grasping with deep reinforcement learning

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AUSTRIAN INSTITUTE
OF TECHNOLOGY

Outline

01 Motivation & Scope

- The grasping problem
- Current task and multi-log objective

02 Simulation & Control

- Crane, grapple, and logs
- From policy actions to motion

03 Learning Formulation

- Observations and grasp detection
- Rewards, safety, and PPO

04 Experiments

- Reward configurations
- Policy behavior and evidence

05 Conclusions & Outlook

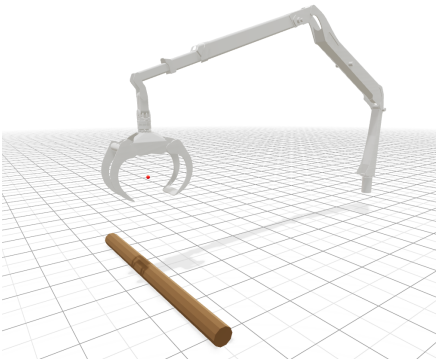
- Main contributions and limitations
- Next steps toward multi-log grasping

Problem $\longrightarrow$ Method

$\longrightarrow$ Evidence $\longrightarrow$ Outlook

The Control Problem

From approaching a log to retaining a physical grasp



Main challenges

- Coordinated motion
- Grasp detection
- Grapple-Log aligment
- Log twisting during grasping
- Straight lifting

Desired sequence

Approach → Align → Close & Clutch → Lift

Project Scope

Single-log development toward multi-log grasping

Current task: one log

- Approach, grasp, and lift
- Contact-based grasp detection
- Reward design and policy training

Multi-log infrastructure

- Configurable log count
- Row piles and retaining barriers
- Fixed-size nearest-logs observations

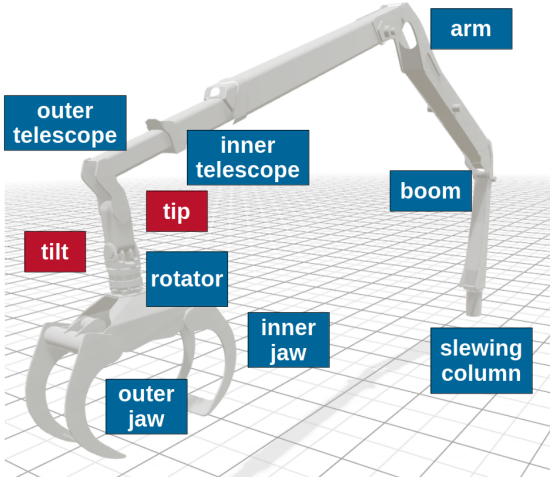
Single-log task $\longrightarrow$ Multi-log grasping

Scope of the current report

Multi-log support is implemented;
Multi-log trainings are yet to be tested.

Forestry Crane Model

Actuated, passive, and mechanically coupled motion



6 actuated joints

- Slewing, boom, arm
- Telescope extension
- Rotator and jaw closure

2 passive joints

Tip and tilt

2 coupled joints

Inner telescope ← outer telescope

Outer jaw ← inner jaw

From Policy Actions to Crane Motion

Six continuous velocity commands

Policy output

Six normalized actions: $\mathbf{a} \in [-1, 1]^6$

Slewing · Boom · Arm
Telescope · Rotator · Jaw

Velocity actuator

$$\tau_j = k_{v,j}\left(v_{d,j} - v_j\right)$$

Velocity-error feedback
subject to actuator effort limits

Joint-specific scaling

$$v_{d,j} = s_j\, a_j$$

Normalized action → desired velocity

Physical response

- Inertia and passive-joint motion
- Contact forces and external load

Policy → Scaling → Actuators → Crane motion

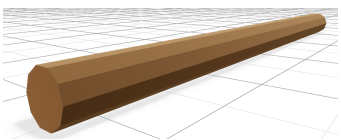
Zero action requests zero velocity—not a return to the home pose.

Log Model & Scene Initialization

Free-body contact, from one log to a pile

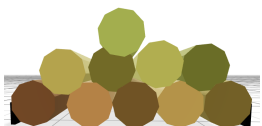
Physical log model

- Decagonal-prism or cilinder mesh
- Free translation and rotation
- Frictional contact



Reusable pile layout

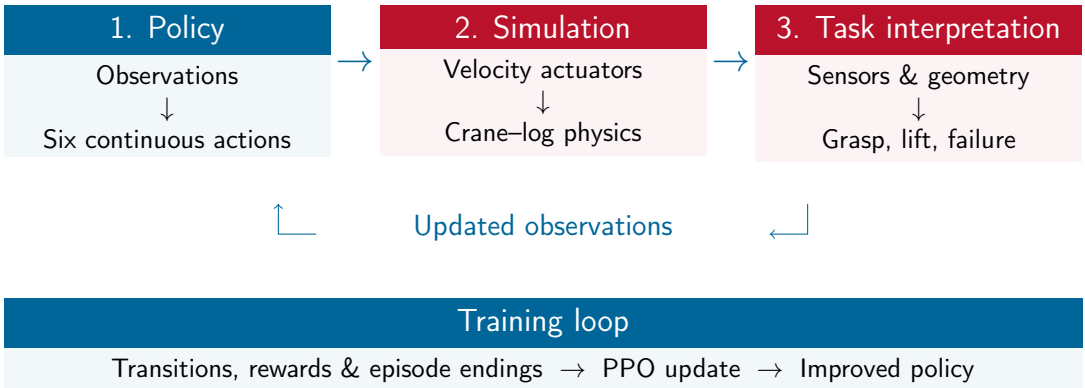
- Centred rows with tapered upper layers
- Retaining barriers for stacked piles



Episode initialization

Place log → Position & yaw jitter → Zero initial velocity → Physics settling

The Closed-Loop Learning Pipeline



What the Policy Observes

Crane & control state

- Joint positions and velocities
- Passive tip and tilt included
- Jaw closure and previous action

Task progress

Grasped count · Hung count
Normalized hanging progress

Log-relative information

- Vector to the remaining pile
- Nearest-log slots:
  - Relative position and heading
  - Motion, horizontalness, lift
  - Validity flag



One real log + nine padded slots

Training actor: selected inputs + noise | Critic: clean inputs

What Counts as a Grasp?

Single-log task: resistance, proximity, and clearance

Shared requirement

Jaw closure resistance + Grasp reference near the log

⇓

Grasped

Shared requirement

+

Log centre above
the log-detach threshold

Hung

Shared requirement

+

Higher centre-height threshold
Both log ends clear of the ground
Approximately vertical grapple

Re-evaluated every step—not permanently latched.

Sensing Engagement & Tracking the Grasp Point

Closure-resistance sensor

Closing actuator force
+ Opposing joint acceleration
+ Jaw inside its safe range

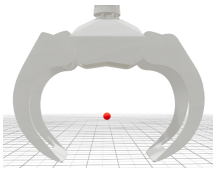
⇓

Mechanical resistance detected

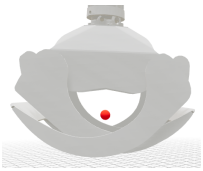
- History retains recent detections
- Opening clears the signal

Dynamic grasp reference

Jaw angle → Reference-point position



Open



Closing

Updates task geometry—not collision geometry.

Resistance sensing + Log proximity → Task-level engagement

Reward Design

Shaping approach, engagement, and stable lifting

Approach & align

- Move toward the log
- Align and centre the grapple

Lift & retain

- Raise and retain the log
- Keep log horizontal, grapple vertical

Engage

- Close near the log
- Establish closure resistance

Discourage unwanted motion

- Abrupt actions and excessive swing
- Log displacement, twisting, and drops

$$r_t = \sum_i w_i \, r_i(s_t, a_t) - \sum_j \lambda_j \, p_j(s_t, a_t)$$

Experiment-specific terms and weights; optional staged schedules.

When Does an Episode End?

Registered episode endings

- Time limit
- Excessive pincer-frame terrain force
- Contact-aware log escape

Penalty $\neq$ termination

Unwanted behavior

$\downarrow$

Reduced return

Episode may continue

Log-escape detection

Touch near the ground $\rightarrow$ Sustained contact loss $\rightarrow$ Escape

Escape requires: near-ground height + sufficient displacement + outward speed + elapsed grace period.

Optional mechanisms: retreat, drop, success, and maximum height.
These affect episodes only when registered.

PPO & Training Setup

Parallel interaction, actor–critic updates

Actor

Noisy observations

↓

MLP: $512 \to 256 \to 128$

↓

Gaussian continuous actions

Critic

Clean observations

↓

MLP: $512 \to 256 \to 128$

↓

State-value estimate

Both networks: ELU activation and observation normalization

Collect → Update → Repeat

B parallel environments $\times$ 24 steps $\to$ PPO: 5 epochs, 4 minibatches

Clipping: 0.2 | Initial learning rate: 3×10^{-4} | $\gamma = 0.99,\ \lambda_{\mathrm{GAE}} = 0.95$

Experimental Configurations

Single-log experiments: planar and full crane motion

Planar: no lifting

- Grasp and straight posture
- No lifting observed

Planar: gentle lifting

- Lift-favoring action ratios
- Staged motion penalties

Full crane motion

- Alignment and motion shaping
- Position and yaw variation

Reported behavior

Grasp without lift | Gentle grasp and lift | Precise grasp; rotation after lift

Planar "Drinking Bird" variant: slewing and rotator blocked.
Qualitative observations—not a controlled ablation or success-rate comparison.

Evaluation Metrics

Episode achievement and pre-hang log motion

Episode achievement

- Grasped at least once
- Hung at least once

Log motion until first hung

- XY error relative to initial position
- Yaw error relative to initial heading
- XY velocity
- Yaw velocity

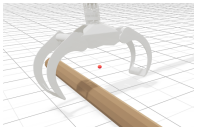
Episode start $\longrightarrow$ Grasp $\longrightarrow$ First hung $\longrightarrow$ Episode end

Two complementary questions

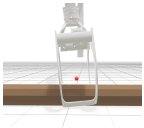
Was grasping / hanging achieved? | How much did the log move before hanging?

Observed Policy Behavior

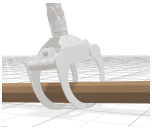
Standard grasp task: `N_rlsum_best_3`, checkpoint 9000



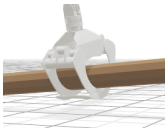
Approach



Alignment



Closing



Closed



Lifted

Episode achievement

Grasped at least once	1
Hung at least once	1

Until first hung

XY error [m]	0.1852
Yaw error [°]	11.8908
XY velocity [m/s]	0.2632
Yaw velocity [°/s]	17.1606

Errors relative to the initial log pose.